INCH-POUND

MIL-T-46316B (AR)
AMENDMENT 4
26 APRIL 2000
SUPERSEDING
AMENDMENT 3
3 JULY 1984

MILITARY SPECIFICATION

TELESCOPE, ELBOW: M118 SERIES AND T176E4

MIL-T 46316B was inactivated after 29 November 1995 for new design

This amendment forms a part of MIL-T 46316B (AR) dated 10 July 1968, and is approved for use by the US Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Paragraph 1.1, delete and substitute as follows:

"1.1 This specification covers similar types of instruments designated as Telescope, Elbow: M118, M118C, M118CA1, and T176E4. These telescopes are basically identical except for differences in reticle patterns."

Paragraph 2.1: Under Drawings, insert, "M118 CA1", after M118C.

Under Packaging Data Sheet, add, "10559855", and "Packaging of Telescope, Elbow: M118CA1".

PAGE 2

Paragraph 3.3.1, delete and substitute as follows:

"3.3.1 Storage temperature – The telescope shall show no evidence of physical failure at room temperature (60° to 90°F) after having been thermally stabilized at -70°F and +160°F."

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Paragraph 3.3.2, delete and substitute as follows:

"3.3.2 Standard ambient temperatures.- Subsequent to meeting the requirements of 3.3.1,the telescope shall meet all the requirements of this specification except 3.3.3 at standard ambient temperature between 60°F and 90°F."

PAGE 3

Paragraph 3.3.3, delete and substitute:

" 3.3.3 Operating temperatures.- The telescope shall show no evidence of physical failure and shall meet the applicable requirements of 3.9 while thermally stabilized at -40°F and +150°F."

Paragraph 3.3.4, insert new paragraph as follows:

"3.3.4 Vibration.- The telescope shall show no evidence of physical failure (loose, missing or broken parts) after having been vibrated at a frequency of 30 cycles per second at an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds. Vibration shall not have caused the eyepiece arm to disengage from the initially engaged position. Subsequent to vibration, the telescope shall meet the requirements of 3.4 to 3.10 inclusive.

Paragraph 3.5, delete and substitute as follows:

"3.5 Sealing.- The telescope shall maintain an internal pressure of dry nitrogen at 5 pounds per square inch (psig) for a minimum period of 2 hours without loss of pressure in excess of 0.2 psig. The interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F."

Paragraph 3.6, second line: delete, "Kingpin", and substitute "tapered".

Paragraph 3.7.1, second line: delete, "the reticle", and substitute, "both reticles."

Paragraph 3.7.5, delete and substitute as follows:

"3.7.5 Boresight error due to reticle rotation.- The line of sight at the boresight point shall not deviate more than 0.25 mil when the reticle is canted a minimum of 5 degrees to the left and a minimum of 5 degrees to the right by means of the cant correction knob."

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* Paragraph 3.8.2, second line: delete, “18 VDC and then 28 VDC”, and substitute “24 VDC”.

Paragraph 3.9, fifth line: delete, “3.3.2”, and substitute, “3.3.3”.

Paragraph 3.9.1, second line: delete, “kingpin”, and substitute, “tapered”.

Paragraph 3.9.3, delete and substitute as follows:

“3.9.3 Mirror, plunger, arm and rheostat.- The level vial mirror (pivotal type only), plunger knob on the reticle shaft lever, eyepiece arm and the rheostat knob shall be mechanically operable when exposed to the operating temperatures of 3.3.3.”

PAGE 5

Paragraph 4.3.2.1, Delete from first sentence: Listed in the list of inspection documents (IEL).”

PAGE 6

Table I, change, “Critical 100%” to “ Critical: None Defined” on next line add “MAJOR 100%” also change, “3.3.3” to “3.3.4”

* Table I characteristics 117 and 118 change test procedure from “4.7.1” to “4.7.2.13”.

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Table II, change, “3.3.2” to “3.3.3”.

PAGE 8

Paragraph 4.6.1.1, change, “3.3.1 and 3.3.2” to “3.3.1, 3.3.2, and 3.3.3” in two places.

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* Paragraph 4.7.1, delete and substitute as follows:

"4.7.1Orientation.- Proper orientation is assured when the telescope is properly mounted and secured to a testing device, the accuracy of which shall be equal to or exceed the accuracy depicted in Drawing F8599924, and when aligned as outlined in set-up instructions outlined on Drawing F8599924. When the telescope is aligned as outlined above, set the mechanical cant index line, and observe that the level vial position is as specified in 3.6. Prior to performing acceptance inspection, remove image tilt by unlocking the eyepiece portion of the telescope and rotating the eyepiece until the longest reticle line is positioned parallel to the comparable collimator target line. Lock in this position. Equipment used for testing the telescope to assure conformance to the requirements in 3.8 shall provide for electrically grounding the power supply used to test the telescope."

Paragraph 4.7.2.2, change, "3.3.3" to 3.3.4" in two places.

Paragraph 4.7.2.5, delete and substitute as follows:

"4.7.2.5 Eyepiece focus.- This test is to be accomplish by placing a white background in front of the telescope 's objective lens. Eyepiece focus is to be measured using a 3 power dioptometer conforming to the accuracies depicted on Drawing 7680631. Adjust the dioptometer eyepiece assembly until a clear, sharp reticle image appears. Next direct the dioptometer's line of sight toward a 500 feet distant target such as a smoke stake, edge of a building, or some other well defined target image. Adjust the objective assembly until parallax is removed. Set the dioptometer scale to zero diopters. The telescope reticle and the image shall remain clear and distinguishable throughout the permissible diopter range specified in paragraph 3.7.6. Repeat this procedure after sliding the alternate reticle (when required) into position."

* Add new paragraph 4.7.2.13 as follows:

"4.7.2.13 Illumination.- Energize the telescope's electrical system and observe that the illumination of the level vial when energized with the voltage specified is in conformance with 3.8.1. With the voltage specified in 3.8.2, observe that the reticle markings are in conformance with 3.8.2 and also observe that a rotation of the rheostat will cause illumination to increase from low to bright."

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Paragraph 5.1, add the following: “10559855 Telescope Elbow M118CA1”.

Add new paragraph 6.3 as follows:

“6.3 Inspection equipment design.

6.3.1 Submission of designs for approval.- Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research Development and Engineering Center, ATTN: AMSTA-AR-QAC-F, Picatinny Arsenal, NJ 07806 -5000. When the contractor submits inspection equipment designs to the government for approval, he shall give the following information in his letter of transmittal:

The contract number.

The contract item (name, model number, etc).

The designs remaining to be submitted and the expected date of submittal.”

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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